enabled drones represent the next evolutionary stage, where advanced computation, ultra-secure communication, and real-time optimization allow UAVs to operate collaboratively and intelligently within dynamic and complex environments. This progression highlights the shift from drones as remotely managed devices to self-adaptive, computing-driven aerial systems capable of supporting high-impact missions across future smart infrastructures.

Table 9.1 Comparison of classical, AI-enhanced, and quantum-enabled drone systems.

Feature	**Classical drones**	**AI-enhanced autonomous drones**	**Quantum-enabled next-generation drones**
Core processing system	Standard microprocessor-based computation	Onboard AI chips/neural processors	Hybrid quantum-classical processors
Decision-making	Human-controlled or semi-automated	Real-time autonomous decision-making	Parallel computation of multiple scenarios simultaneously
Navigation capability	GPS-dependent; limited obstacle avoidance	SLAM-based navigation and adaptive path planning	Hyper-optimized routing and navigation in complex + dynamic environments
Perception/object recognition	Basic camera feed; manual interpretation	Computer vision + deep learning for object detection	Quantum-enhanced pattern recognition for high-complexity visual analysis